

摘 要

随着数字图像技术广泛应用以及互联网技术的不断发展和普及应用，数字图像的安全性保护和版权保护问题变得迫在眉睫。为解决这一问题，盲数字水印嵌入与提取技术因其不可见性和难以被破坏等特点而成为一种有效的数字版权保护方式。基于 Arnold 置乱和 ZUC 加密的 DWT 域数字水印算法作为一种组合式创新的算法能够广泛地应用的数字水印技术，并且已经受到了广泛的研究并且实现应用。

本文针对当今数字图像中存在的安全性问题，并结合数字水印技术的应用，改进得出了一种基于二维 Arnold 映射和 ZUC 加密的 DWT 域数字水印算法方案。并且本文采用 Arnold 映射和 ZUC 加密算法，在 DWT 域中嵌入数字水印，以提高数字水印的安全性并实现数字图像的版权保护和安全性保护，从而增强数字水印的鲁棒性。本文将首先介绍数字水印技术的研究背景和发展现状，重点着眼于 DWT 域数字水印技术以及 Arnold 置乱和 ZUC 加密技术的原理，并探讨它们在数字水印中的应用。随后，本文将详细阐述基于 Arnold 置乱和 ZUC 加密的 DWT 域数字水印算法的原理和实现步骤。本算法通过采用离散小波变换将数字水印嵌入到低频系数中的方式和手段，并通过 Arnold 置乱和 ZUC 加密的方式来保证数字水印的安全性。最后，本文将通过对多幅测试图像进行实验验证和对比分析，以验证基于 Arnold 置乱和 ZUC 加密的 DWT 域数字水印算法的有效性和可行性。同时，对实验结果进行分析以验证整个算法系统的安全性和鲁棒性。

关键词：离散小波域变换；数字水印；Arnold 置乱；ZUC 加密；水印抗性测试

ABSTRACT

With the wide application of digital images and the development and popularization of Internet technology, the security protection and copyright protection of digital images have become an urgent problem to be solved. Blind digital watermarking technology has become an effective means of digital copyright protection because of its invisibility and not easy to be destroyed. DWT domain digital watermarking algorithm based on Arnold scrambling and ZUC encryption can be widely used as a combined innovative algorithm, and has received extensive research and attention.

This paper proposes a digital watermarking algorithm in the DWT domain, utilizing Arnold scrambling and ZUC encryption to address the security issues in current digital images. The algorithm enhances the security of digital watermarking by incorporating these techniques, while embedding the digital watermark in the DWT domain to achieve copyright protection and security for digital images, thereby improving the robustness of the digital watermarking.

This paper first introduces the research background and development status of digital watermarking technology, focusing on the principles of DWT domain digital watermarking technology and Arnold scrambling and ZUC encryption technology and their application in digital watermarking. Then, the principle and implementation steps of DWT domain digital watermarking algorithm based on Arnold scrambling and ZUC encryption proposed in this paper are elaborated. This algorithm embeds the digital watermark into the low-frequency coefficient by using discrete wavelet transform, and ensures the security of the digital watermark through Arnold scrambling and ZUC encryption. Finally, through experimental verification and comparative analysis of multiple test images, the effectiveness and feasibility of the algorithm are verified, and the experimental results are analyzed to verify the safety and robustness of the overall algorithm system.

Keywords: Discrete wavelet domain transform; Digital watermarks; Arnold messed up; ZUC encryption; Watermark resistance test